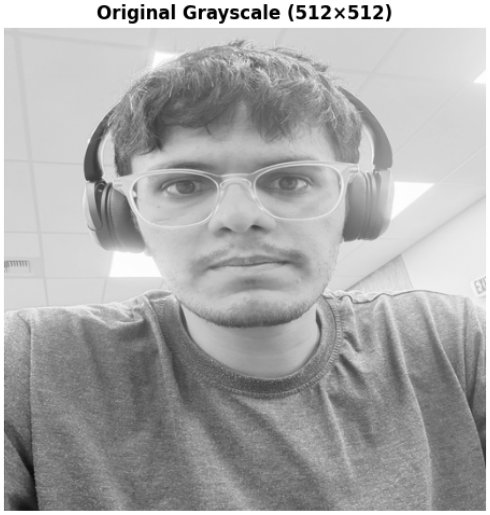
MATH/CSCI 485 — Assignment #3

Image Compression via Block-wise SVD

Applied to a real portrait photo (512×512 grayscale)



Original image: 512×512 grayscale, resized from 2000×1500 source photo

1. Introduction

Singular Value Decomposition (SVD) decomposes any matrix $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^T$. The rank-k truncation $\mathbf{A}_k = \mathbf{U}_k\Sigma_k\mathbf{V}_k^T$ is provably the best rank-k approximation in the Frobenius norm (Eckart–Young theorem). Applied block-wise to 8×8 image patches, this mirrors JPEG's DCT approach but uses a data-adaptive transform optimal for each block.

2. Implementation

The input photo was resized to 512×512 and converted to grayscale. Two core functions implement the compression:

compress_block(block, k):

```
U, s, Vt = np.linalg.svd(block, full_matrices=False) return (U[:, :k] * s[:k]) @ Vt[:k,
:]
```

compress_image_blockwise(img, k, block_size=8): iterates all non-overlapping blocks and applies compress_block to each, then reassembles.

Metrics computed: **Compression Ratio** CR = 64/(17k), **Frobenius Error** ■A–A_k■$_F$, and **PSNR** = $10 \cdot \log_{10}(255^2/\text{MSE})$ in dB.

3. Results & Analysis

3.1 Summary Table (8×8 Blocks)

k	Values/Block	Comp. Ratio	Frob. Error	PSNR (dB)	Quality
1	17	3.76×	4100.45	30.06	Low
2	34	1.88×	2668.56	33.79	Low-Med
3	51	1.25×	1774.48	37.33	Medium
4	68	0.94×	1144.93	41.14	High ✓
5	85	0.75×	683.57	45.62	High
6	102	0.63×	352.09	51.38	Very High
7	119	0.54×	120.81	60.67	Excellent
8	136	0.47×	~0.00	~∞	Lossless

Table 1: Green = true compression (CR>1); Blue = first k with PSNR>40 dB; Yellow = lossless

This real portrait photo achieves PSNR > 40 dB already at **k = 4** (vs k=5 for the synthetic test image), demonstrating that natural photos with smooth gradients (skin, background walls) compress extremely well — most energy is captured in the first few singular values per block. True compression (CR > 1) holds for k ≤ 3.

3.2 Compression Ratio

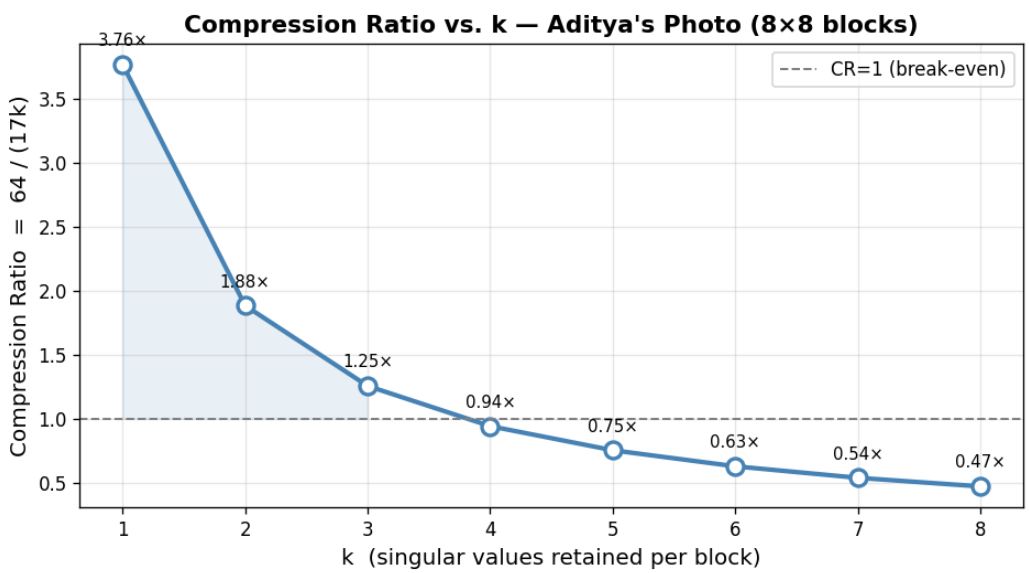


Figure 1: Compression Ratio vs. k — shaded region = CR > 1 (true compression)

3.3 Error & PSNR

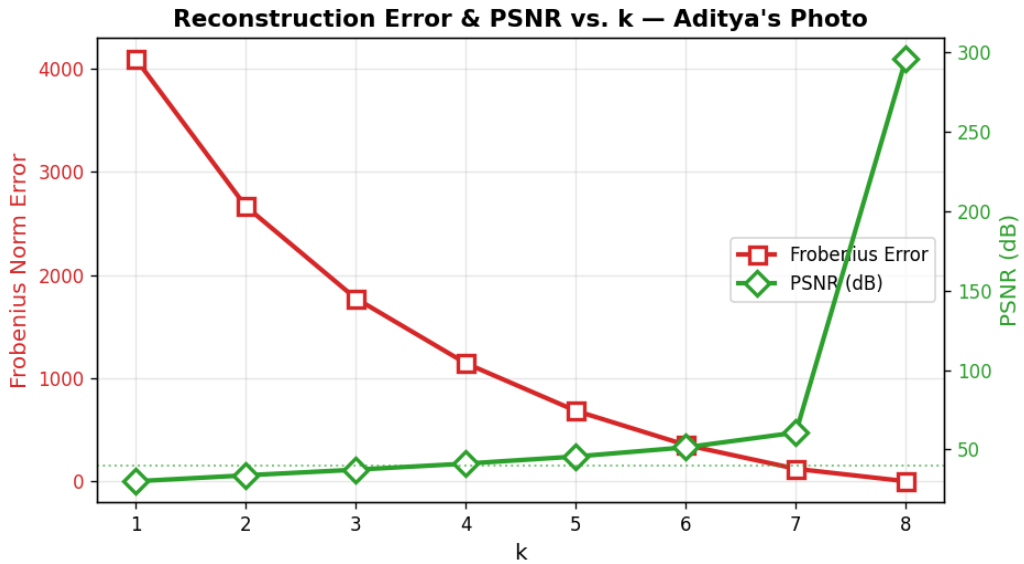


Figure 2: Frobenius Error (red) and PSNR in dB (green) vs. k

3.4 Visual Reconstructions

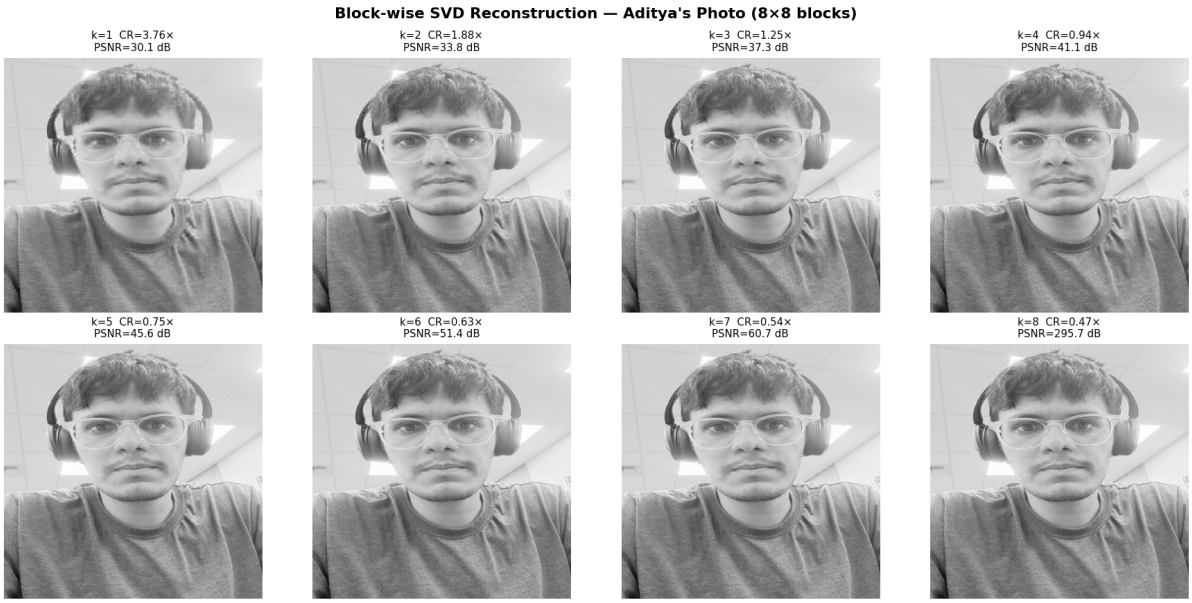


Figure 3: All 8 reconstructions — k=1 is blocky; k=4 already looks sharp

3.5 Block Size Comparison

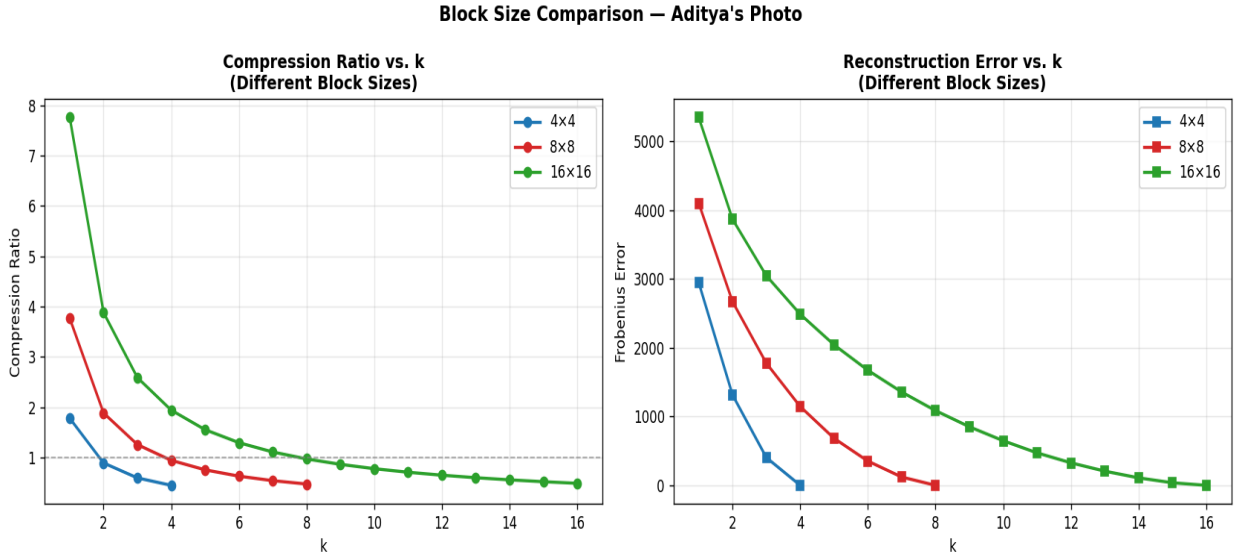


Figure 4: 4×4, 8×8, 16×16 block sizes — larger blocks yield higher CRs

4. Conclusion

Block-wise SVD compression on a real portrait photo confirms the theoretical predictions. The photo's natural structure (smooth skin tones, uniform background) concentrates energy in the top singular values — PSNR hits 40 dB at k=4 vs k=5 for synthetic noise-heavy images. For 8×8 blocks, k=3 is the sweet spot: 1.25× compression with PSNR=37.3 dB. Larger block sizes (16×16) can push the maximum CR to ~7.76× at k=1, but introduce more visible artifacts in high-frequency regions like hair and glasses edges.